4 – bit bidirectional shift register with parallel load

I M.Sc., Physics

By

Mrs. Pushpalatha Gracelin

Bidirectional shift register with parallel load

- Shift registers are used for converting

  # serial data to parallel data and

  # parallel data to serial data.

- If we access all the flip-flop outputs of a shift register, then data entered serially by shifting can be taken out in parallel.
- If a parallel load capability is added to a shift register, then data entered parallel can be taken out in serial.
- Shift registers provide input and output terminals for parallel transfer.
- They have both shift-right and shift-left capabilities.

Most general shift register has all the functions given below:

- A **clear** control to clear the register to 0.
- A **CP input** for clock pulses to synchronize all operations.
- A **shift-right** control to enable the shift-right operation.

  Serial input and output lines associated with the shift-right.
- A **shift-left** control to enable the shift-left operation.

  Serial input and output lines associated with the shift-left.
- A **parallel-load** control to enable parallel transfer.

  n input lines associated with the parallel transfer.
- n parallel output lines.
- A **control state** that leaves the information in the register

unchanged even though clock pulses are continuously applied.

Bidirectional shift register :

A register capable of shifting both right and left.

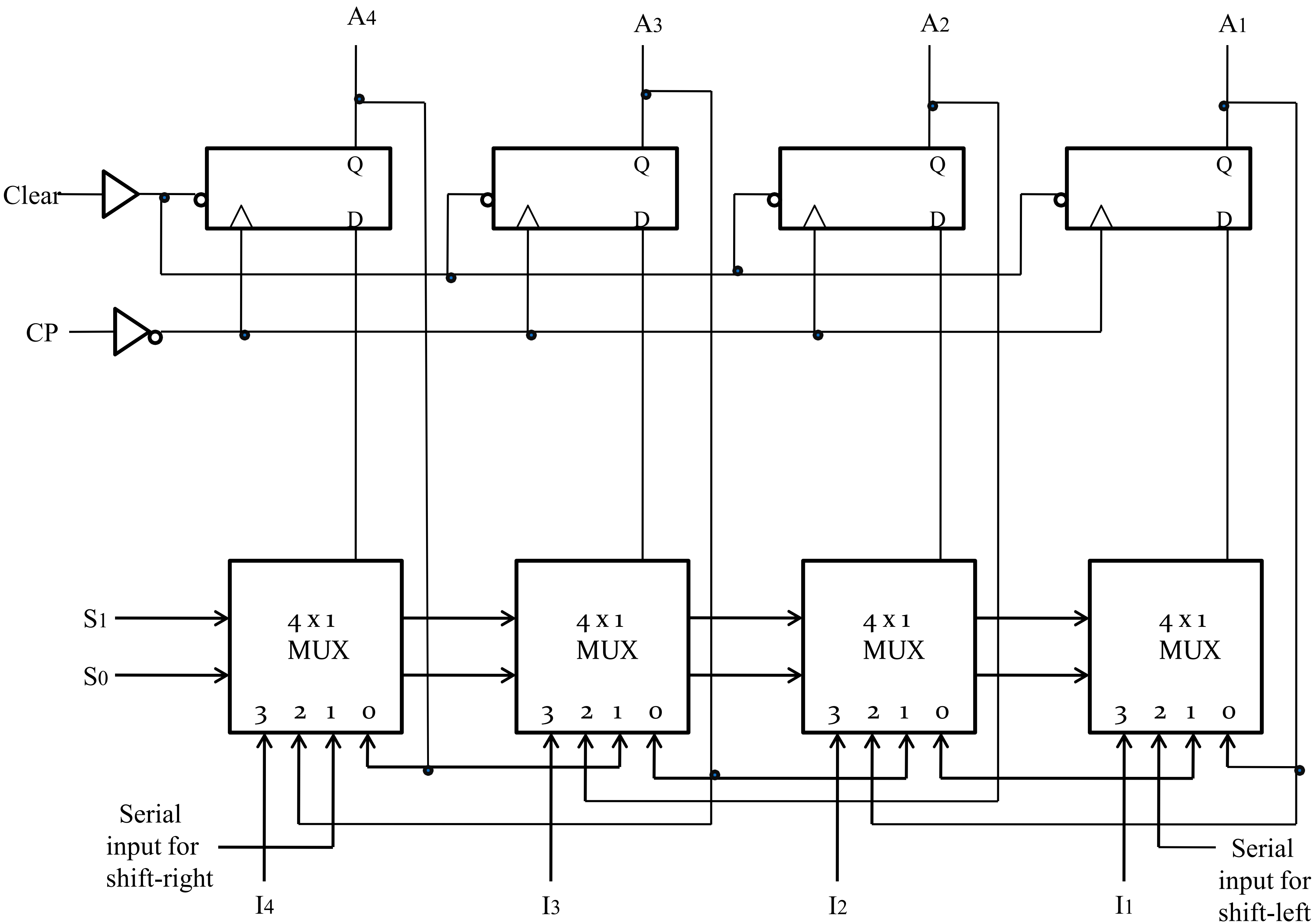
Unidirectional shift register :

Register which shift in only one direction.

Shift register with parallel load :

The register which has both shift and parallel-load capabilities.

4 – bit bidirectional shift register with parallel load



Shift register consists of

Four D flip-flops

RS flip-flop used when an inverter is inserted between S and R

terminals.

Four Multiplexers (MUX)

Four multiplexers have two common selection variables, S1 and S0.

When $S_1S_0 = 00$, input 0 is selected

When $S_1S_0 = 01$, input 1 is selected

S_1 and S_0 inputs control the mode of operation of the register.

Function table for the Register

Mode Control S_1	S_0	Register Operation
0	0	No change
0	1	Shift right
1	0	Shift left
1	1	Parallel load

When $S_1S_0 = 00$,

- The present value of the register is applied to D inputs of flip-flops.
- This condition forms a path from the output of each flip-flop into the input of the same flip-flop.
- Next clock pulse transfers into each flip-flop the binary value it held before, and no change of state occurs.

When $S_1S_0 = 01$,

- Terminals 1 of the multiplexer inputs have a path to D inputs of flip-flops.
- This causes shift-right operation, with serial input transferred into flip-flop A_4.

When $S_1S_0 = 10$,

Shift-left operation occurs, with serial input going into flip-flop A_1.

When $S_1S_0 = 11$,

Binary information on parallel input lines is transferred into the register continuously during the next clock pulse.

Bidirectional shift register with parallel load is a **general purpose register**.

It performs three operations:

i. Shift-left

ii. Shift-right and

iii. Parallel load.

THANK YOU